

Diagnostic Evaluation Report

1. Framework Information

- **Framework:** X-RAG Diagnostic Framework (v1.0)
- **Trace ID:** `015f991c-babb-402c-b205-76e8f36f1848`
- **Timestamp:** 2026-09-05T14:29:41.174327Z
- **Status:** COMPLETED

2. Executive Summary

- **Question:** What penalty does the BNS prescribe for a company's failure to file annual financial statements with the R
- **Answer:** I do not have enough information to answer this.
- **Health Score:** 0.64
- **Primary Issue:** UNKNOWN
- **Claims Supported:** 1 / 1
- **Summary:** Analysis completed.

3. Pipeline Overview

Stage	Status	Confidence	Observation
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CORPUS	PASS	0.9	Corpus contains sufficient relevant content, evidenced by at least one supported claim.
RETRIEVER	PASS	0.95	At least one retrieved chunk provides supporting evidence for a verified claim.
GENERATOR	PASS	0.95	All generated claims are fully supported by evidence.
GROUNDING	PASS	0.9	No claims are contradicted by retrieved evidence.

Efficiency: 1/6 chunks used (17%)

4. Evaluation Metrics

- **Retrieved Chunks:** 6
- **Verified Claims:** 1 (Supported: 1, Partial: 0, Unsupported: 0, Contradicted: 0)
- **Grounding Score:** 1.00
- **Evidence Coverage:** 0.17
- **Average Entailment:** 0.99
- **Latencies:** Retrieval (2439.5ms) | Generation (10418.7ms) | Verification (5194.6ms)

5. RAGAS Metrics

- **Faithfulness:** 1.0000
- **Answer Relevancy:** 0.3889
- **Context Precision:** 0.0000
- **Context Relevancy:** 0.0000
- **Context Recall:** N/A
- **Answer Similarity:** N/A
- **Answer Correctness:** N/A

Answer Correctness (Claim Recall)

Not computed for this trace.

6. Evidence Analysis

- **[9a02a76d-8c11-4116-a504-a7df3db4baec]** (SUPPORTED): I do not have enough information to answer this.
- *Evidence (Chunk f0949e68-3b60-4c7c-bf7d-add5bfd13d91):* The entries are relevant, but are not sufficient, without other evidence, to prove the debt.

7. Root Cause Analysis

- **Primary Cause:** UNKNOWN
- **Confidence:** 0.93
- **Reasoning:**
 - Starting root cause analysis for trace 015f991c-babb-402c-b205-76e8f36f1848.
 - CORPUS passed (confidence=0.90).
 - RETRIEVER passed (confidence=0.95).
 - Skipping CHUNKING: no observable state in matrix.
 - GENERATOR passed (confidence=0.95).
 - GROUNDING passed (confidence=0.90).
 - No failures detected. Pipeline is healthy or all evidence is UNKNOWN.

8. Corrective Action Plan

- **[INFORMATIONAL] Review Retrieval Top-K / Reranker Cutoff**
- Description: Retrieval efficiency is low (17%); consider reducing top-k or tightening the reranker score cutoff.
- Expected Improvement: Reduced prompt size and lower latency/cost with no loss of verified support. (Metric: Chunk utilization)

9. Evidence Traceability

- **Primary Cause:** UNKNOWN
- **Derived From:** RootCauseAnalysis
- **Pipeline Stage:** UNKNOWN
- **Supporting Claims:** 9a02a76d-8c11-4116-a504-a7df3db4baec
- **Supporting Chunks:** f0949e68-3b60-4c7c-bf7d-add5bfd13d91

10. Overall Assessment

- **Major Strength:** RETRIEVER is healthy (confidence=95%): At least one retrieved chunk provides supporting evidence for
- **Major Weakness:** UNKNOWN
- **Next Priority:** Review Retrieval Top-K / Reranker Cutoff
- **Recommendation:** Pipeline requires targeted improvements to the primary failure stage.

11. Framework Footer

- **Generated By:** X-RAG Diagnostic Framework
- **Artifact Version:** 1.0

APPENDIX

Artifacts Generated

✓ RAGTrace

artifacts/rag_traces/015f991c-babb-402c-b205-76e8f36f1848.json

✓ ClaimSet

artifacts/claim_sets/015f991c-babb-402c-b205-76e8f36f1848.json

✓ VerificationResults

artifacts/verification/015f991c-babb-402c-b205-76e8f36f1848.json

✓ PipelineStateMatrix

artifacts/pipeline_state_matrix/015f991c-babb-402c-b205-76e8f36f1848.json

✓ RootCauseAnalysis

artifacts/root_cause_analysis/015f991c-babb-402c-b205-76e8f36f1848.json

✓ CorrectiveActionPlan

artifacts/corrective_action_plan/015f991c-babb-402c-b205-76e8f36f1848.json

✓ DiagnosticEvaluationReport

artifacts/reports/015f991c-babb-402c-b205-76e8f36f1848.json